Palatial Pools

OpenHands Technical Architecture

1. System Overview and Capabilities

OpenHands operates as an advanced autonomous agent framework within the Palatial Pools infrastructure. It provides multi-step code orchestration, enabling complex development and administrative tasks to be executed autonomously. The system is designed to handle file creation and editing, execute terminal commands, browse the web, and manage sandboxed environments for secure execution.

By leveraging the OpenAI o3 model, OpenHands possesses reasoning capabilities that allow it to understand complex requirements and execute them through a series of logical steps. This makes it an invaluable tool for system maintenance, script generation, and deep technical troubleshooting.

2. Infrastructure and Deployment

The OpenHands instance is deployed as a containerized application on the primary Palatial Pools Virtual Private Server (VPS). This deployment strategy ensures isolation while maintaining the necessary access to system resources for administrative tasks.

Server Configuration Details

The system is configured using Docker, providing a consistent and reproducible environment. The deployment utilizes specific container images to ensure compatibility and stability.

Component	Configuration
Primary Image	`ghcr.io/all-hands-ai/openhands:0.15`
Runtime Environment	`ghcr.io/all-hands-ai/runtime:0.15-nikola`
Internal Port	`3000`
Exposed Port	`3005` (Mapped as `3005:3000`)
Access URL	`autopilot.palatialpools.com.au`

Volume Mappings

To function effectively, OpenHands requires specific access to the host system. The volume mappings are configured to balance functionality with security.

The Docker socket (`/var/run/docker.sock`) is mounted into the container. This critical configuration allows OpenHands to spawn its own sandboxed environments, which is essential for safely executing untrusted code or isolating complex tasks. Additionally, a local data directory (`./data:/app/data`) is mounted to provide a persistent workspace for the agent to read and write files during its operation.

3. Integration and Security

OpenHands is integrated into the Palatial Pools web stack through a secure reverse proxy configuration. This ensures that the powerful capabilities of the agent are strictly controlled and monitored.

Network and Authentication

The service is exposed via an Nginx reverse proxy, which handles SSL termination and routes traffic from `autopilot.palatialpools.com.au` to the internal port 3005.

Security is enforced at the Nginx layer using Basic Authentication. Access is restricted to authorized personnel only. The authentication requires the username `kieran` and the designated administrative password. This layer of security is paramount, given the system's ability to execute arbitrary code and commands.

4. Known Limitations and Constraints

While highly capable, the current OpenHands deployment has specific operational constraints that must be considered during use.

State Management

The most significant limitation is the lack of persistent state across conversational sessions. The memory resets completely between sessions. Therefore, any context, previous instructions, or ongoing task knowledge must be re-established if a session is interrupted or a new one is initiated.

Performance and Resource Usage

OpenHands can exhibit slower performance when handling complex, multi-file tasks. The reasoning process of the underlying model, combined with the overhead of managing sandboxed environments, can lead to extended execution times.

Furthermore, the utilization of the OpenAI o3 model incurs high API costs. This financial constraint necessitates judicious use of the system, prioritizing tasks where its advanced reasoning provides substantial value over simpler, automated scripts.

5. Troubleshooting Guide

When encountering issues with the OpenHands deployment, the following steps can help identify and resolve common problems.

If the web interface is inaccessible, first verify that the Nginx reverse proxy is running and correctly configured to route traffic to port 3005. Check the Nginx error logs for any authentication failures or routing errors.

If the interface loads but the agent fails to execute commands or create sandboxes, verify the Docker socket volume mapping. OpenHands requires access to `/var/run/docker.sock` to function correctly. Ensure the permissions on the host socket allow the container to interact with the Docker daemon.

For issues related to file persistence or missing workspaces, confirm that the `./data` directory is correctly mapped and has the appropriate read and write permissions for the container user.

© 2026 Palatial Pool Services Pty Ltd. Internal Technical Documentation.